

Our project aimed to create a fire detection and suppression system that met all the design requirements. These requirements included detecting and suppressing the fire within one minute, staying under a \$50 budget, operating everything within the structure of the model, and making it function independently without any human interaction once the fire starts. Our final system used an Arduino kit, a smoke detector, a small water pump, and tubing placed along the ceiling of the model to work as a water sprinkler system.

STEM principles were foundational in addressing each of our design requirements. To meet the requirements used some physics, since smoke rises we knew to make sure to place the smoke detector on the ceiling. Once smoke was detected, the arduino would trigger the water pump, activating the suppression system. We also needed to know some programming to be able to code the arduino.

Meeting the \$50 budget limit required careful analysis of the cost of the materials we needed. We made sure to use low cost materials to try and stay under the budget. This required math and engineering skills to ensure an efficient design and material management.

The system's ability to work fully within the strcutures's area and rely on an external power source showed more applied engineering and physics. We had to carefully plan how to place the pump, tubing, and Arduino without going out of the model's dimensions. This needed attention to space everything out. Physics concepts like the voltage, current, and energy transfer were also essential to make sure the system worked well.

Another requirement was that the system function without any human intervention after the fire had started. We used the Arduino to monitor the smoke detector and activate the pump automatically when smoke was detected. This helped us to better understand sensor data handling and digital signal processing which are key elements of computer science and systems design.

Unfortunately, during testing, our smoke detector did not work as we expected. While the pump and tubing setup were in place and ready, we could not fully test the system's reaction to smoke in the final test. Despite this set back, our design decisions were reviewed by engineering professionals and our teacher. This feedback showed us the strengths of our design and also offered feedback for improvements.

The experts we talked to had backgrounds in engineering and fire protection, making them well qualified to review the STEM content of our project.

Based on their feedback, we were able to fix very important stuff such as getting a bigger tank of water to keep the water flowing and to not get a whole smoke detector and take it apart.

In conclusion, our project showed a wide range of STEM concepts, including things such as physics, math, and engineering design. Even though we had some technical issues during testing, our solution demonstrated a clear use of these concepts to real world constraints, and

expert feedback provided us with a path to improve in the future. With continued work, our design has high potential to become a fully functional fire suppression system,